

From Spec to Visuals: Rapid API Testing with Postman

Sakshi Nasha



Agenda

- Introduction to yours Truly
- Challenges
- Tricks & Tips
- Import OpenAPI/Swagger into Postman
- Visualize API Responses
- Automate Auth with Token Handling
- Supercharge with Scripting & Variables
- Bonus



\$whoami

- Developer 
- OpenSearch Ambassador
- AWS community Builder 
- Innovator : Hackathons
- Athlete at heart :    





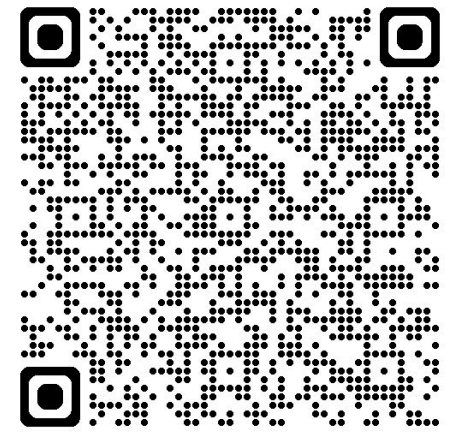
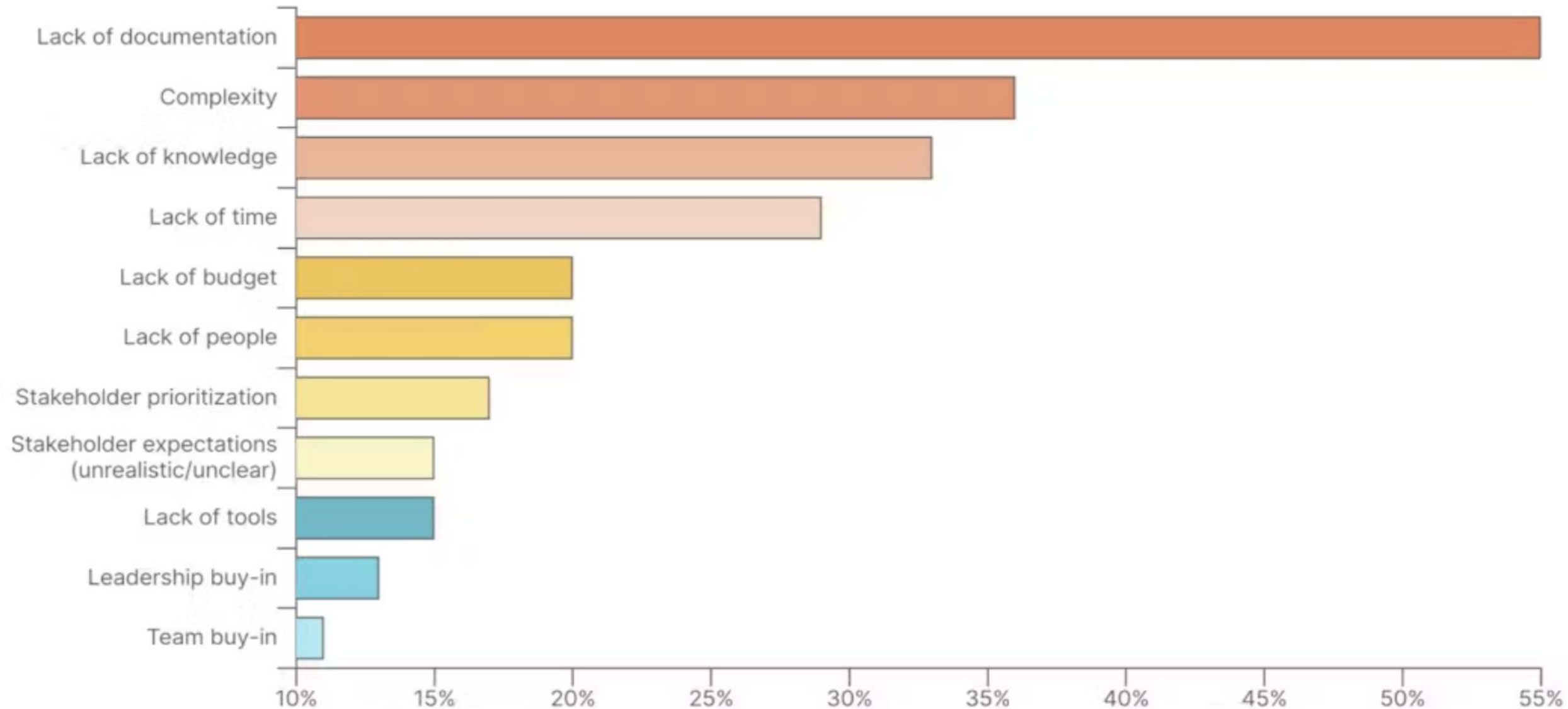
Challenges faced today

- **APIs are everywhere, but testing them is still clunky.**
 - **Complex API Architectures** (microservices and hybrid models)
 - **Manually reading Swagger docs**
 - **Bad Documentation**
 - **Copy-pasting cURLs**
 - **Poor visibility in responses**



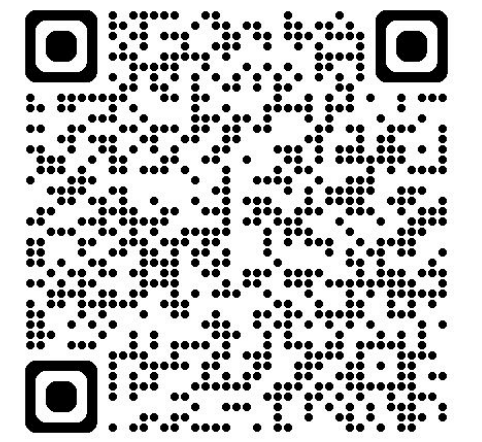
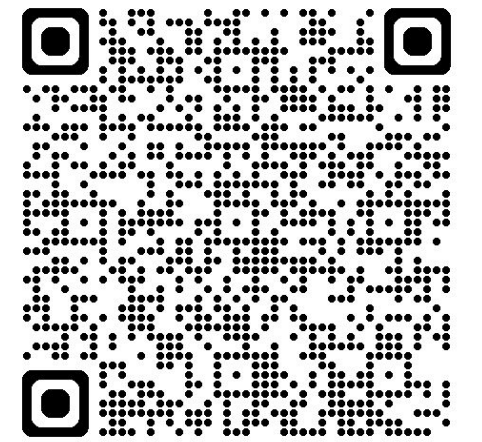
2021 State of the API Report

Obstacles to consuming APIs



Challenges faced today

- **Huge financial losses:** Poor API performance costs the industry ~US\$92B/year
- **Documentation is a major blocker:** ~50-60% of developers cite lack of documentation as the top obstacle to using or integrating APIs.
- **Manual debugging & testing dominates:** Lots of time wasted in manual steps rather than automated or spec-driven workflows.



#1 : Import OpenAPI/Swagger into Postman

- Just drag-and-drop or use the Import URL feature
- Example : <https://petstore.swagger.io/>
- How to use in Postman:

Import → Link → Paste above URL

Also, Upload a .yaml or .json spec

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Alta.RVLT.API

IJS

OpenSearch

RecoveryVaultAPIv208-Authorization

RecoveryVaultAPIv208-Credentials

RecoveryVaultAPIv208-GMS

RecoveryVaultAPIv208-Provisioning

RecoveryVaultAPIv208-Telemetry

RecoveryVaultAPIv208-Token

Tool Rental API

GET Status

GET Tools

GET Assertions on Objects

GET Single Tool

POST Get token

POST Create Order

GET Get all Orders

GET Single order

PATCH Update Order

DEL Delete Order

GET Check if Order is deleted

GET Automate API

HTTP

Tool Rental API / Status

GET

{{toolrentalBaseURL}}/status?foo=boo

Send

Params

Authorization

Headers (6)

Body

Pre-request Script

Tests

Settings

1

pm.test("The status is UP", ()=>{

2

pm.response.to.have.status(200);

3

});

Choose how to import your API

☒ Postman Collection

☐ Swagger Petstore

☐ OpenAPI 2.0 with a Postman Collection

Work with the definition file along with an exclusive collection

View Import Settings

Back

Import



Click Send to get a response

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NewImport

Swagger Petstore

- pet
 - {petId}
 - uploadImage
 - POST uploads an image
 - GET Find pet by ID
 - POST Updates a pet in the sto...
 - DEL Deletes a pet
 - findByStatus
 - findByTags
 - POST Add a new pet to the store
 - PUT Update an existing pet
- store
 - inventory
 - GET Returns pet inventories ...
 - order
- user
 - createWithList
 - {username}
 - login
 - GET Logs user into the system
 - logout
 - createWithArray
 - POST Create user

HTTP

Swagger Petstore / user / login / Logs user into the system

Save

Send

GET

{{baseUrl}}/user/login?username=<string>&password=<string>

ParamsAuthorizationHeaders (7)BodyPre-request ScriptTestsSettingsCookies

Query Params

☒	Key	Value	Description	Bulk Edit
☒	username	<string>	(Required) The user name for login	
☒	password	<string>	(Required) The password for login in clear ...	
	Key	Value	Description	

BodyCookiesHeaders (10)Test Results

Status: 200 OKTime: 37 msSize: 471 BSave as example

PrettyRawPreviewVisualizeJSON

```
1 {
2   "code": 200,
3   "type": "unknown",
4   "message": "logged in user session:1758394098227"
5 }
```

OnlineFind and replaceConsoleImport CompletePostbotRunnerStart ProxyCookiesTrash



#2 : Visualize API Responses

- Emphasize “Client-ready visuals without frontend code”
- `pm.visualizer.set()` snippet in the test tab
- Example : GET

<https://petstore.swagger.io/v2/pet/findByStatus>

- Send request -> output is a JSON doc
- Add Visualization Script in the "Tests" Tab
- View the Output

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Swagger Petstore

pet

{petId}

uploadImage

GET Find pet by ID

POST Updates a pet in the sto...

DEL Deletes a pet

findByStatus

GET Finds Pets by status

findByTags

POST Add a new pet to the store

PUT Update an existing pet

store

inventory

GET Returns pet inventories ...

order

user

createWithList

{username}

login

GET Logs user into the system

logout

createWithArray

POST Create user

Swagger Petstore / pet / findByStatus / Finds Pets by status

Save

Send

GET

{{baseUrl}}/pet/findByStatus?status=available&status=available

Params

Authorization

Headers (7)

Body

Pre-request Script

Tests

Settings

```
1 const pets = pm.response.json();
2
3 const tableHtml = `
4   <table border="1" style="border-collapse: collapse; width: 100%;">
5     <tr>
6       <th>ID</th>
7       <th>Name</th>
8       <th>Status</th>
9     </tr>
10    ${pets.map(pet => `
11      <tr>
12        <td>${pet.id}</td>
13        <td>${pet.name}</td>
14        <td>${pet.status}</td>
15      </tr>
16    `).join('')}
17  </table>
18 `;
19
20 pm.visualizer.set(tableHtml);
21
```

Body

Cookies

Headers (8)

Test Results

Status: 200 OK

Time: 37 ms

Size: 32.7 KB

Save as example

Online

Find and replace

Console

Import Complete

Postbot

Runner

Start Proxy

Cookies

Trash

Collections

Environments

Flows

Monitors

History

Swagger Petstore

pet

{petId}

uploadImage

GET Find pet by ID

POST Updates a pet in the sto...

DEL Deletes a pet

findByStatus

GET Finds Pets by status

findByTags

POST Add a new pet to the store

PUT Update an existing pet

store

inventory

GET Returns pet inventories ...

order

user

createWithList

{username}

login

GET Logs user into the system

logout

createWithArray

POST Create user

HTTP

Swagger Petstore / pet / findByStatus / Finds Pets by status

GET

{{baseUrl}}/pet/findByStatus?status=available&status=available

Params

Authorization

Headers (7)

Body

Pre-request Script

Tests

Settings

☒	status	available	(Required) Status values
☒	status	available	(Required) Status values
	Key	Value	Description

Body

Cookies

Headers (8)

Test Results

Status: 200 OK

Time: 37 ms

Size: 32.7 KB

Pretty

Raw

Preview

Visualize

JSON

```

1  [
2    {
3      "id": 9223372000001084490,
4      "category": {
5        "id": 0,
6        "name": "string"
7      },
8      "name": "doggie",
9      "photoUrls": [
10       "string"
11     ],
12     "tags": [
13       {
14         "id": 0,
15         "name": "string"
16       }
17     ],
18     "status": "available"
19   },
20   {
21     "id": 9223372000001084491,

```

Online

Find and replace

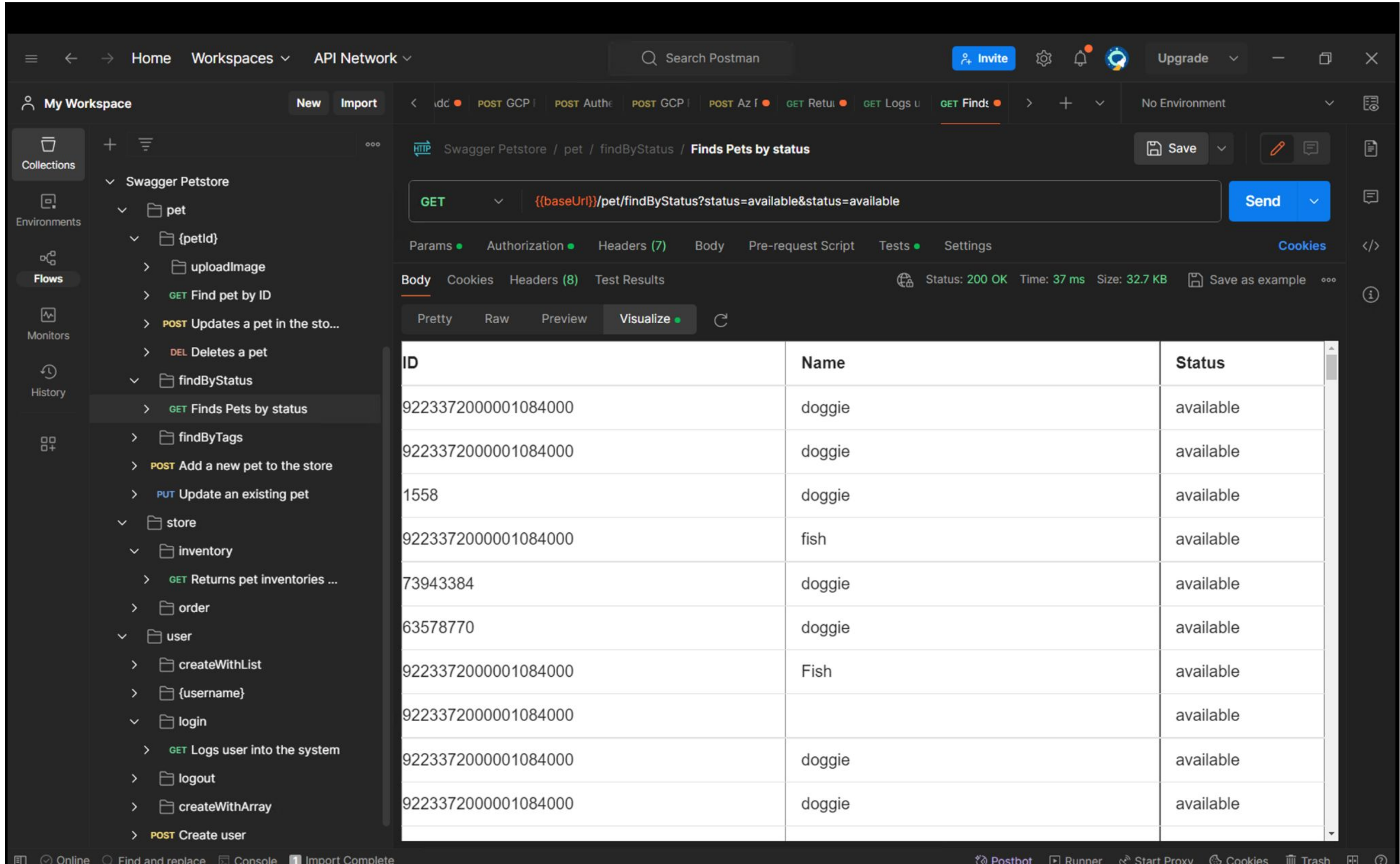
Console

1 Import Complete

Postbot

Runner

Start Pr



Body Cookies (1) Headers (7) Test Results (0/3) Security

{ } JSON Preview Visualize

```

1 {
2   "args":
3   "cit
4   "ram
5 },
6 "headers
7 "hos
8 "x-r
9 "con
10 "x-forwarded-proto": "https",
11 "x-forwarded-port": "443",

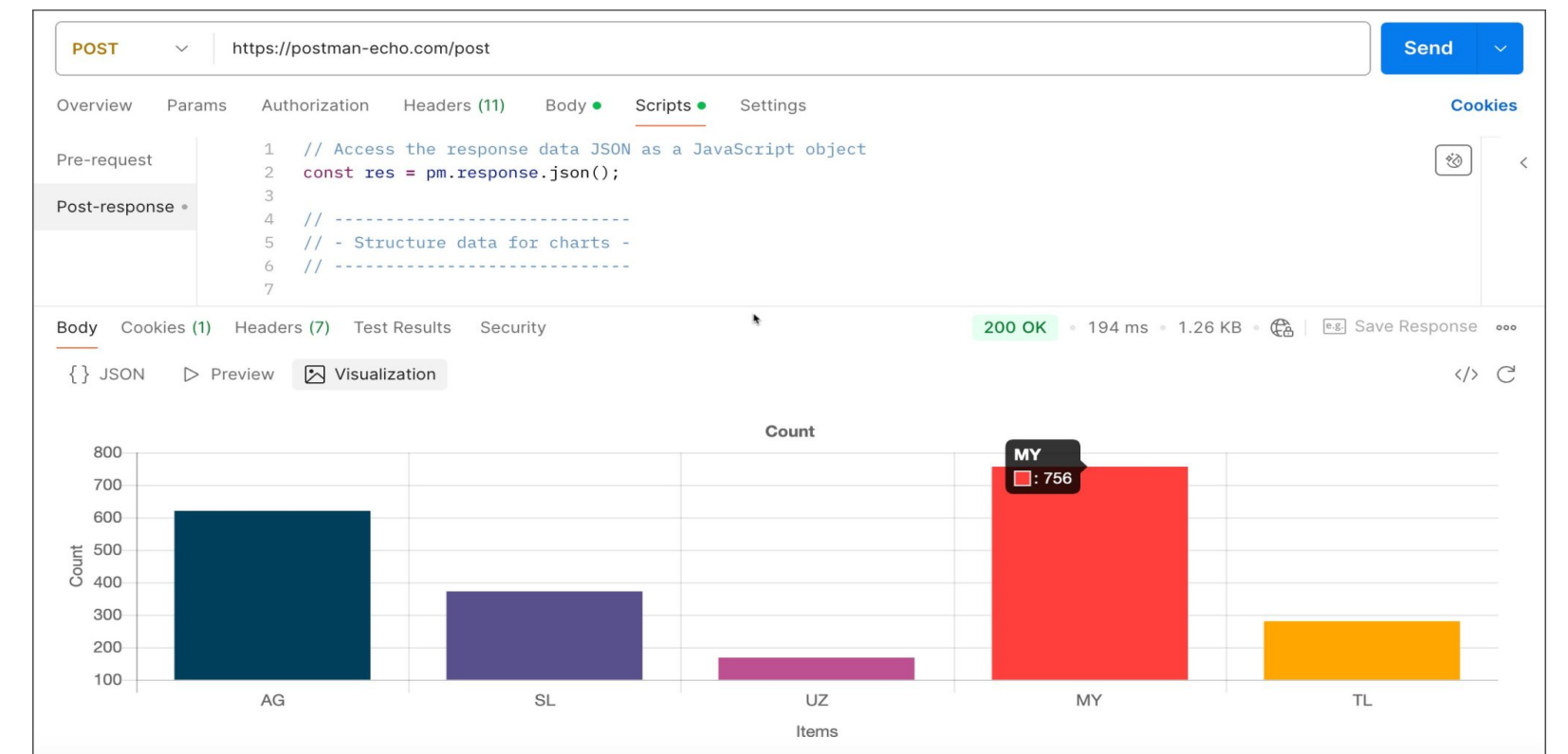
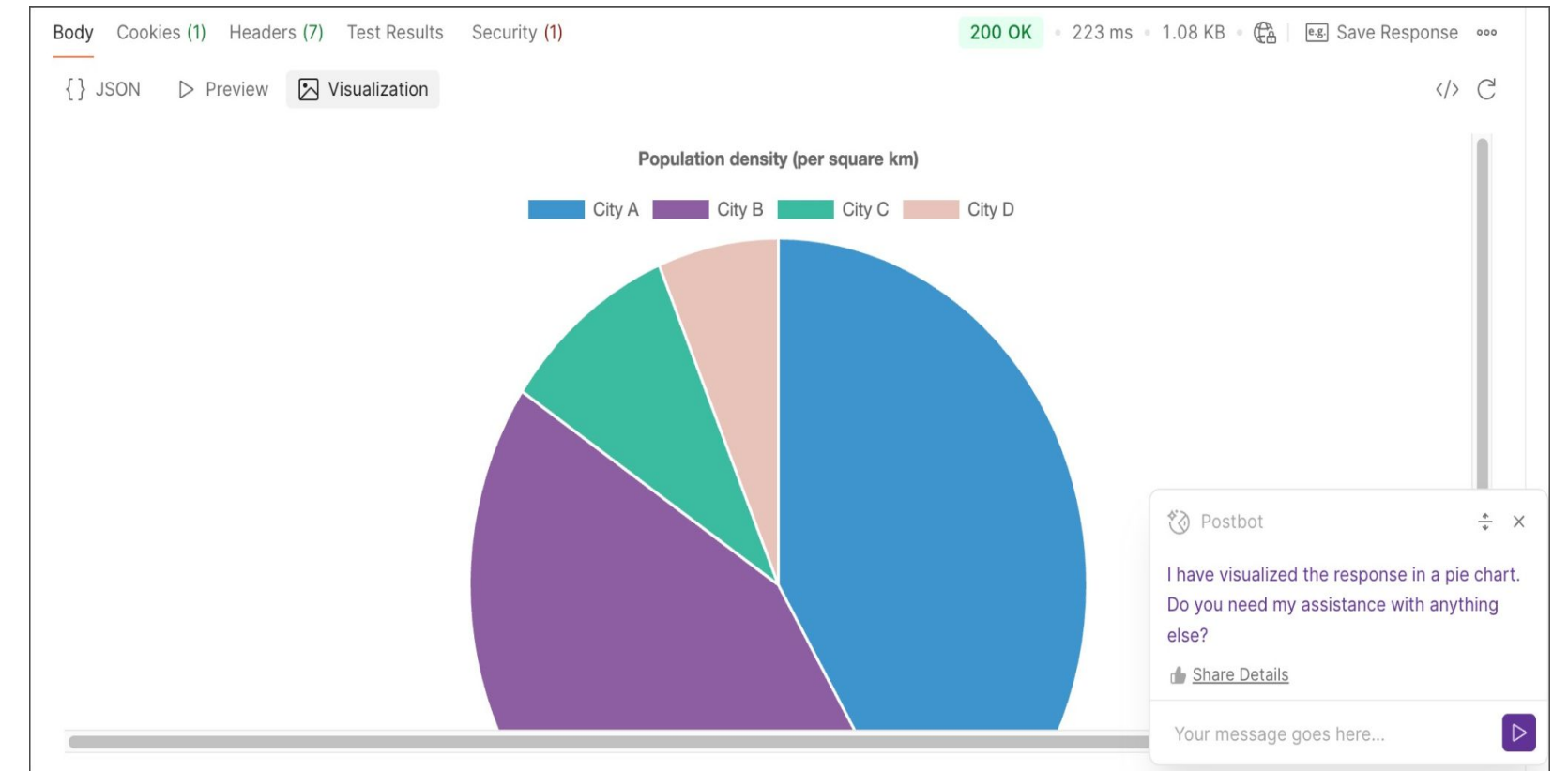
```

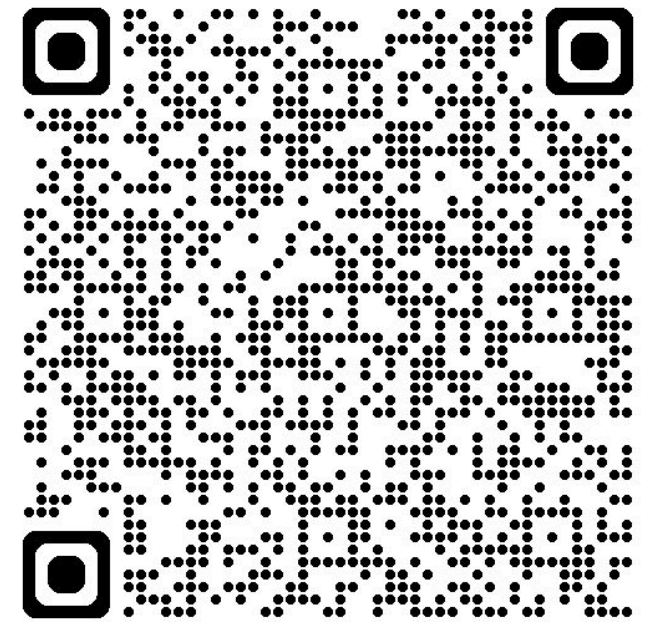
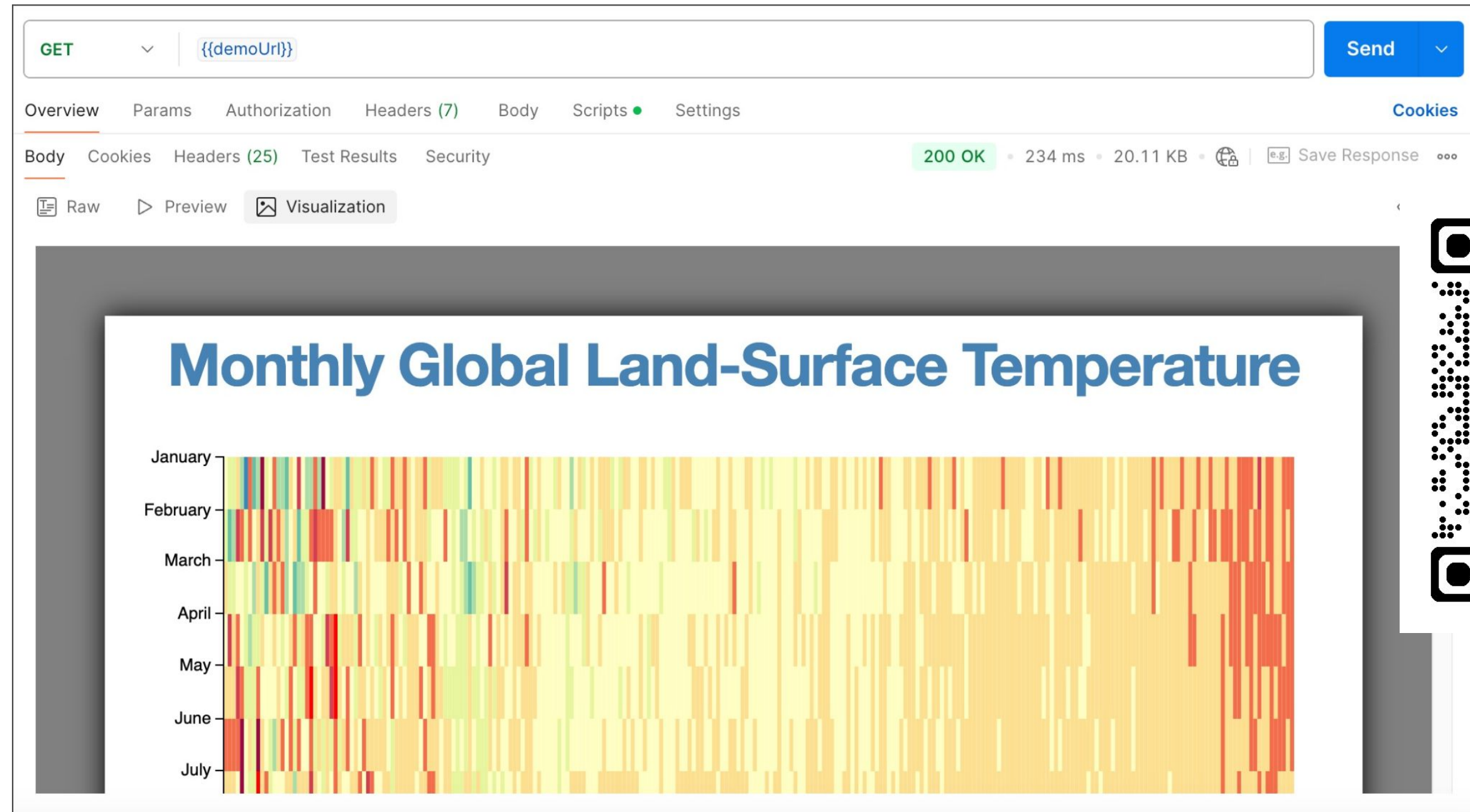
Table

Linear Chart

Bar Graph

Set up with a prompt





GET

https://earthquake.usgs.gov/fdsnws/event/1/query?format=geojson&starttime=2019-10-13&endtime=2019-10-16

Send

Overview

Params

Authorization

Headers (7)

Body

Scripts

Settings

Cookies

Body

Cookies

Headers (22)

Test Results

Security (1)

200 OK

1.03 s

121.77 KB

e.g. Save Response

{ } JSON

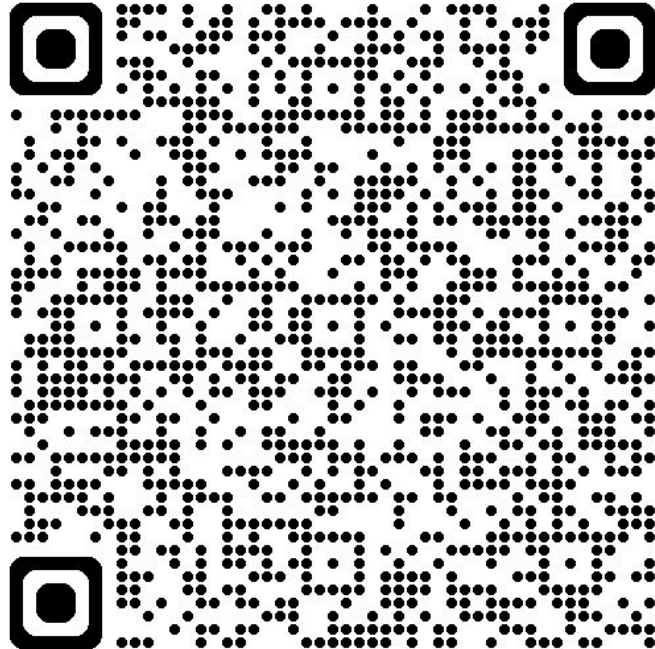
Preview

Visualization

Map of Earthquake Occurances over Three Days

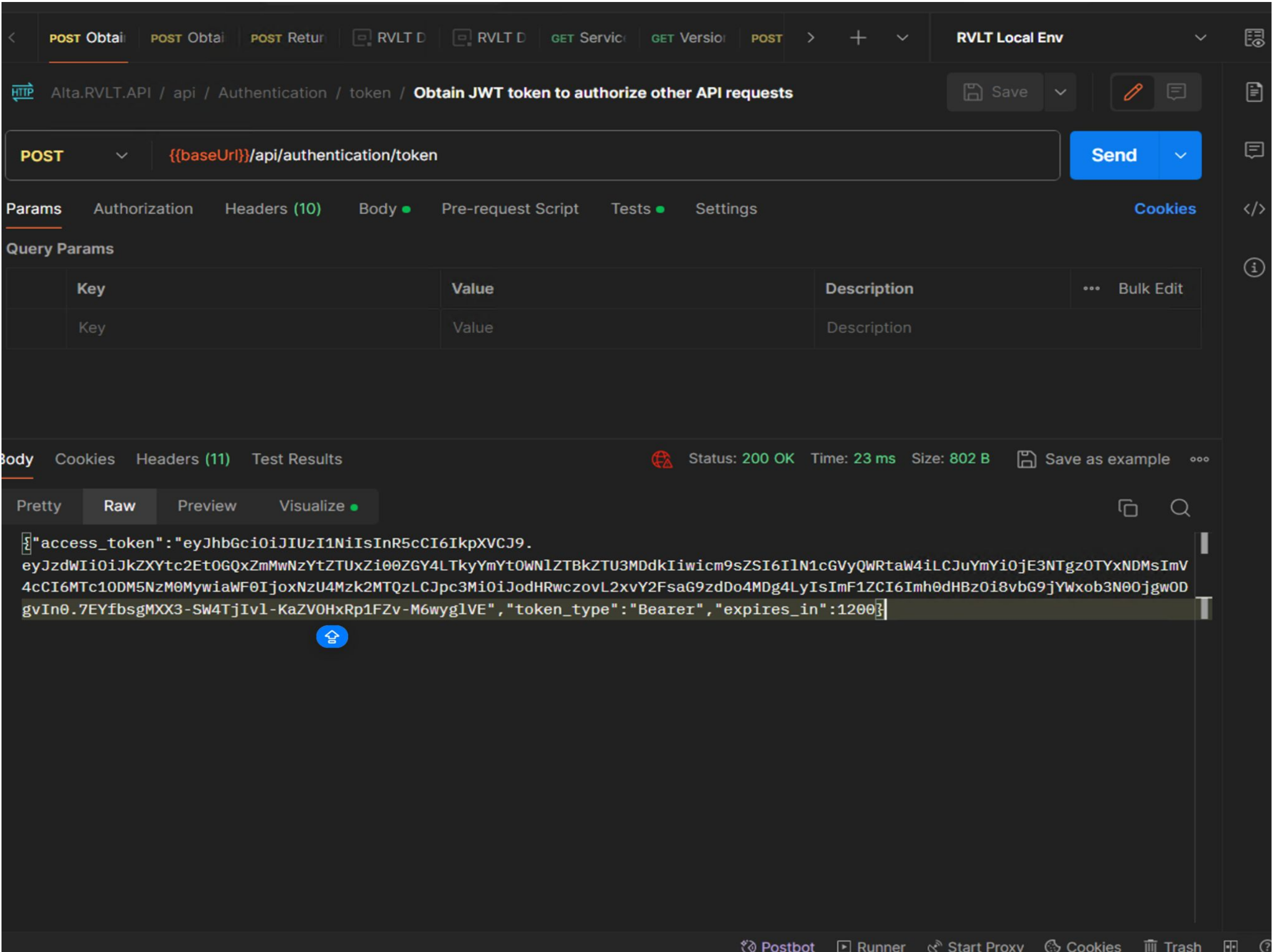


Draw mouse to zoom in onto section. Double click to



#3 : Automate Auth with Token Handling

- Pain: Copy-pasting tokens is slow and error-prone
- Solution: Use Postman environments + scripts
 - Store token from login response
 - Inject Authorization: Bearer {{auth_token}} in headers
 - Refresh token in pre-request scripts



<

POST Obtain

POST Obtain

POST Return

RVLT D

RVLT D

GET Service

GET Version

POST

>

+

▼

RVLT Local Env

▼

⋮

Alta.RVLT.API / api / Authentication / token / Obtain JWT token to authorize other API requests

Save ▼

✎

💬

⋮

POST ▼ {{baseUrl}}/api/authentication/token

Send ▼

Params Authorization Headers (10) Body ● Pre-request Script Tests ● Settings Cookies </>

19 ,

20

21 function constructVisualizerPayload() {

22 | return {response: data};

23 }

24

25 pm.visualizer.set(template, constructVisualizerPayload());

26

27

Body Cookies Headers (11) Test Results

🌐 Status: 200 OK Time: 23 ms Size: 802 B Save as example ⋮

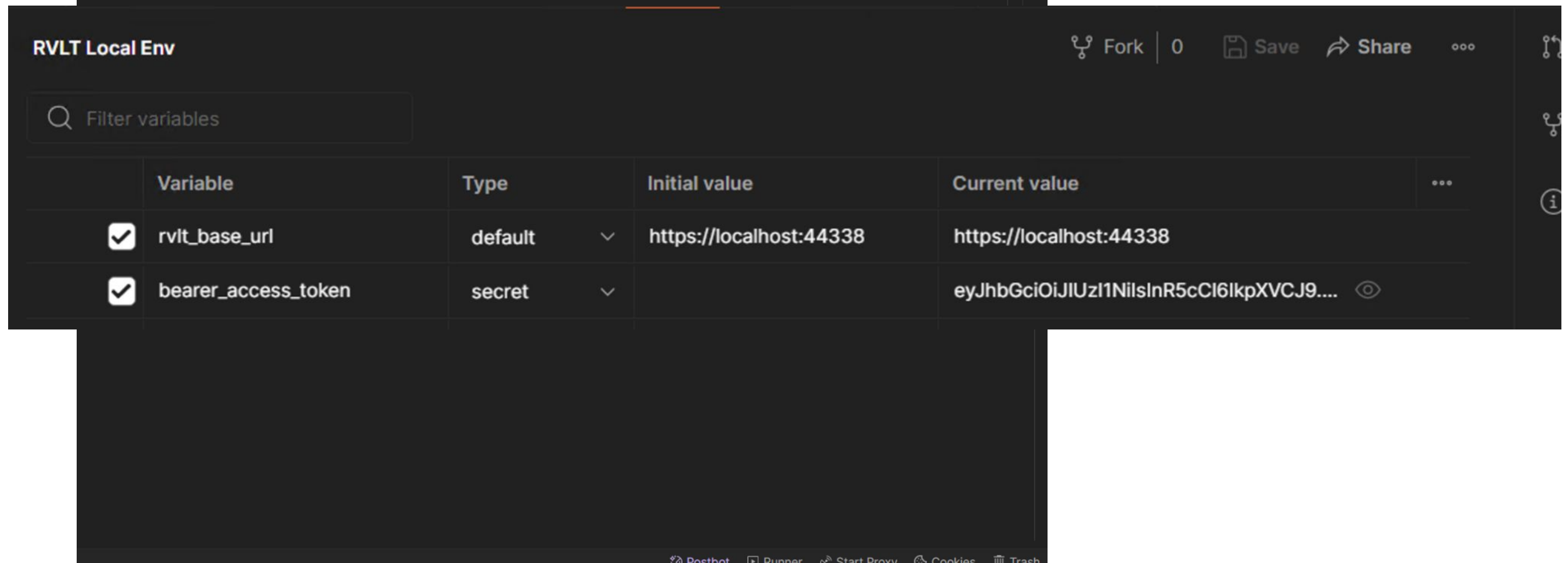
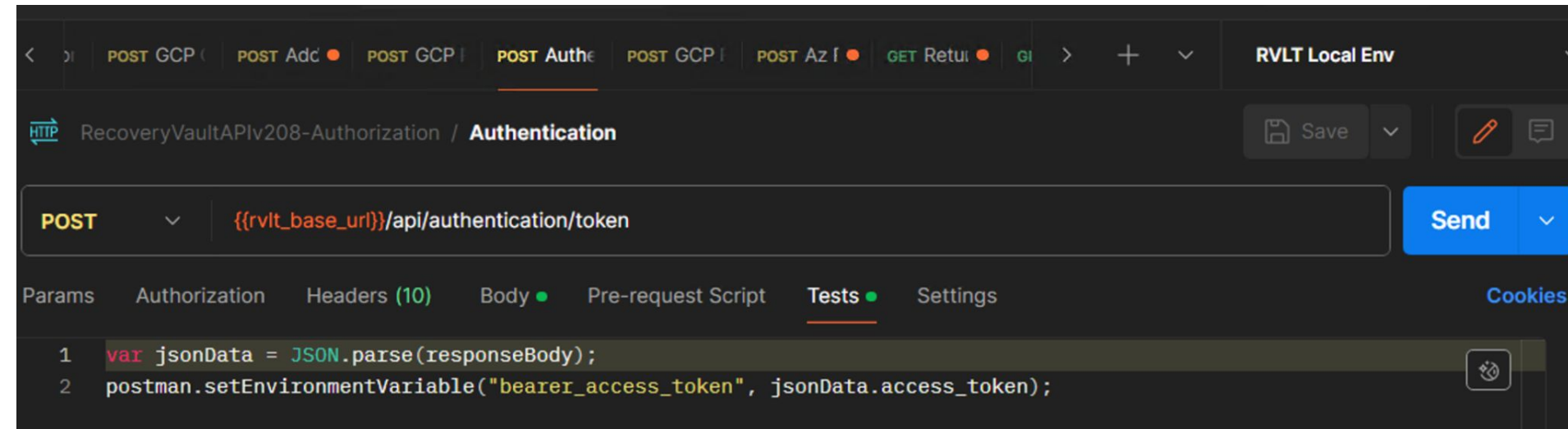
Pretty Raw Preview Visualize ● ↻

Token Details

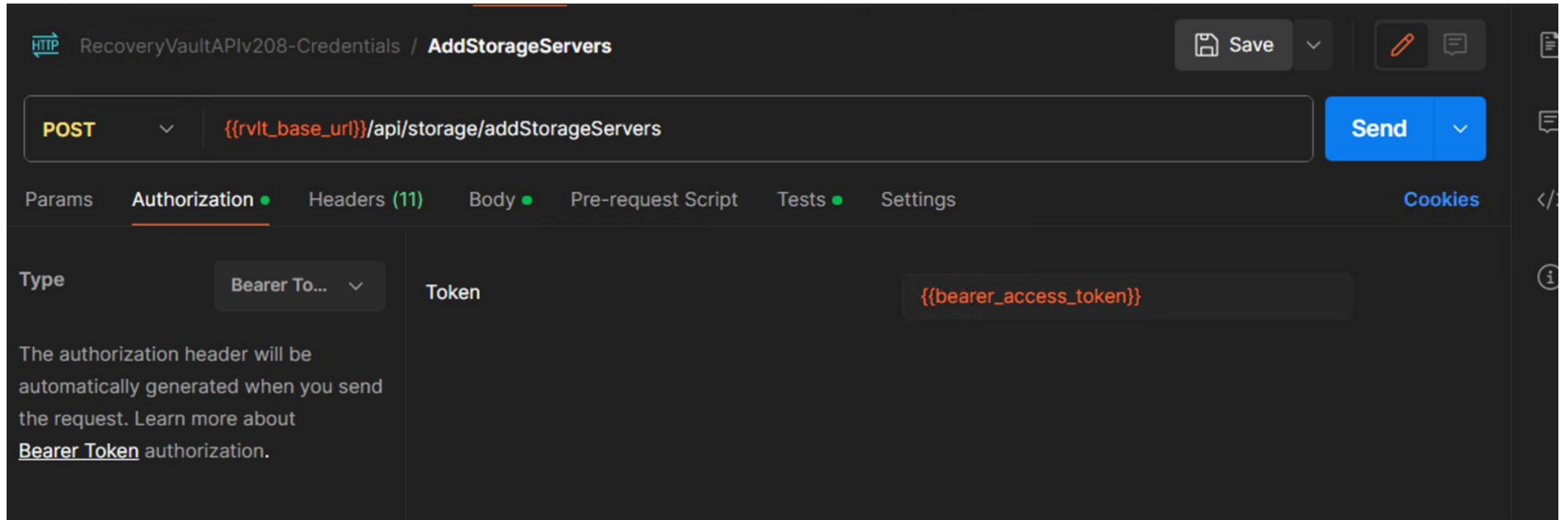
Key	Value
Access Token	eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJkZXZtc2EtOGQxZmMwNzYtZTUxZi00ZGY4LTkyYmYtOWNIZTtSW4TjIvIiKaZVOHxRp1FZv-M6wyglVE
Token Type	Bearer
Expires In (seconds)	1200

Postbot Runner Start Proxy Cookies Trash

Set env Variable of “bearer_access_token”



Using env Variable “ bearer_access_token” in another api Call



The screenshot shows the Postman interface for an API call. The top bar indicates the collection 'RecoveryVaultAPIv208-Credentials' and the endpoint 'AddStorageServers'. The method is 'POST' and the URL is '{{rvlt_base_url}}/api/storage/addStorageServers'. The 'Authorization' tab is selected, showing 'Type' as 'Bearer To...' and 'Token' as '{{bearer_access_token}}'. A note explains that the authorization header will be automatically generated when the request is sent.

HTTP RecoveryVaultAPIv208-Credentials / **AddStorageServers** Save

POST {{rvlt_base_url}}/api/storage/addStorageServers Send

Params **Authorization** Headers (11) Body Pre-request Script Tests Settings Cookies

Type Bearer To... Token {{bearer_access_token}}

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

#4 : Supercharge with Scripting & Variables

- Use `pm.environment.set()`, `tests[]`, `pre-request`
- Examples:
 - Chain requests
 - Parse response and set next input
 - Fail tests on bad responses
- Benefits: faster debugging, fewer clicks



Request 1-> Create a pet, store its ID using pm.environment.set()

HTTP

Swagger Petstore / pet / Add a new pet to the store

Save

POST

▼

{{baseUrl}}/pet

Send

▼

Params

Authorization ●

Headers (9)

Body ●

Pre-request Script

Tests ●

Settings

Cookies

</>

Pet API

Fork | 0

Save

Share

...

Filter variables

	Variable	Type	Initial value	Current value	...
☒	pet_id	default		9223372036854775285	
	Add new variable				

```
2  "id": 9223372036854775285,
3  "name": "demo-daisy",
4  "photoUrls": [
5    "https://example.com/bruno.jpg"
6  ],
7  "tags": [],
8  "status": "available"
9  }
```

Request 2 : Get Pet by ID

 Swagger Petstore / pet / {petId} / Find pet by ID

GET

{{baseUrl}}/pet/{{pet_id}}

Send

Params

Authorization

Headers (8)

Body

Pre-request Script

Tests

Settings

Cookies

```
1 let pet = pm.response.json();
2
3 pm.test("Status code is 200", function () {
4   pm.response.to.have.status(200);
5 });
6
7 pm.test("Pet has a name", function () {
8   pm.expect(pet).to.have.property("name");
9   pm.expect(pet.name).to.be.a("string").and.not.empty;
10 });
11
```

Body

Cookies

Headers (8)

Test Results (2/2)

 Status: 200 OK Time: 153 ms Size: 449 B  Save as example 

 Perform API tests faster with templates for integration testing, regression testing, and more.

View Templates



All

Passed

Skipped

Failed



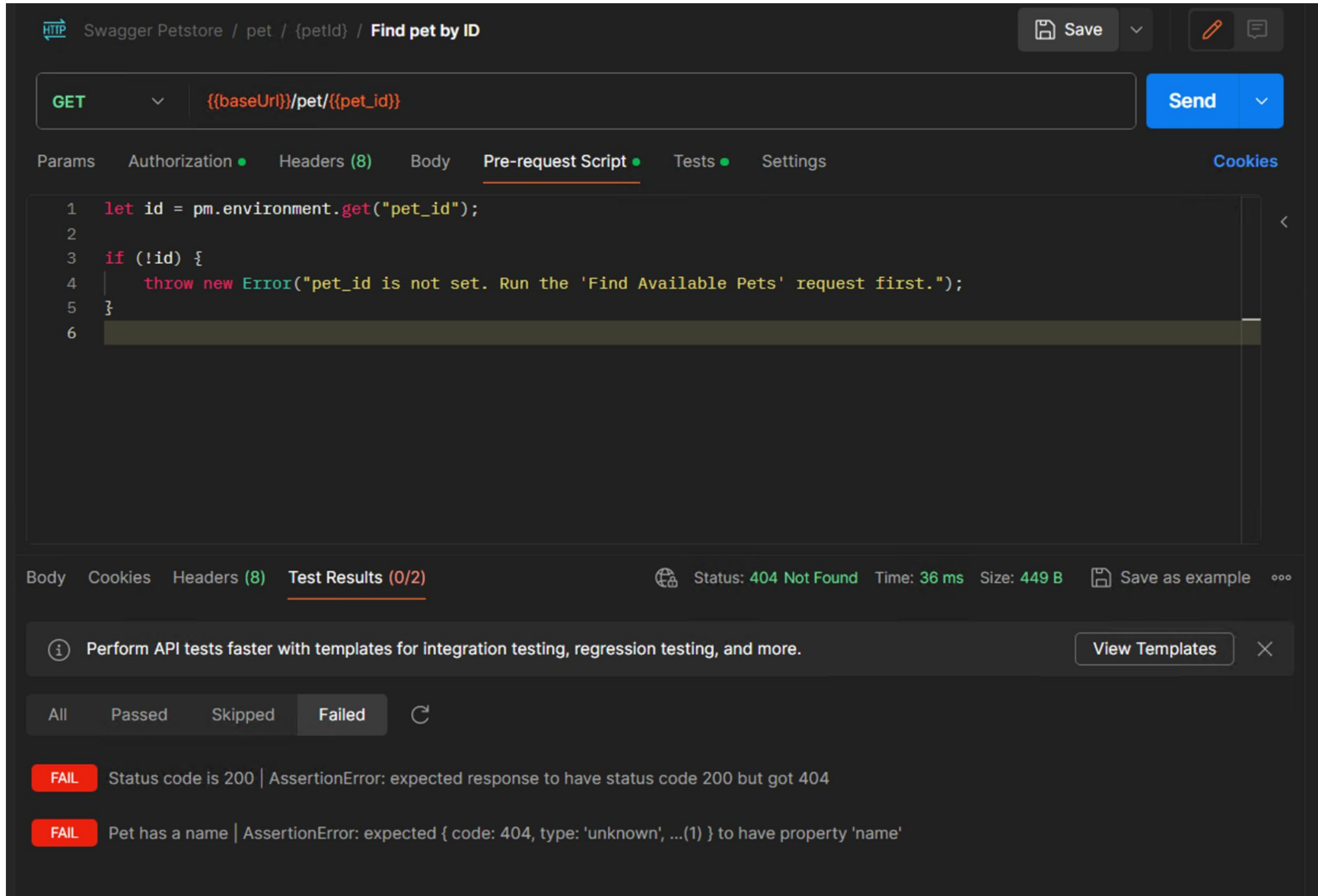
PASS

Status code is 200

PASS

Pet has a name

Request 2 : Pre Request Script



Swagger Petstore / pet / {petId} / Find pet by ID

GET {{baseUrl}}/pet/{{pet_id}}

Send

Params Authorization Headers (8) Body Pre-request Script Tests Settings Cookies

```
1 let id = pm.environment.get("pet_id");
2
3 if (!id) {
4   throw new Error("pet_id is not set. Run the 'Find Available Pets' request first.");
5 }
6
```

Body Cookies Headers (8) Test Results (0/2) Status: 404 Not Found Time: 36 ms Size: 449 B Save as example

Perform API tests faster with templates for integration testing, regression testing, and more. View Templates

All Passed Skipped Failed

FAIL Status code is 200 | AssertionError: expected response to have status code 200 but got 404

FAIL Pet has a name | AssertionError: expected { code: 404, type: 'unknown', ...(1) } to have property 'name'



#4 : Supercharge with Scripting & Variables

- Instead of copying values between requests, I chain them using environment variables. I create a pet, store its ID using `pm.environment.set()`, and use that in the next call automatically.
- This reduces clicks, avoids mistakes, and makes my test flow much cleaner.
- I also used PreRequest Script -> if id is not found throw error

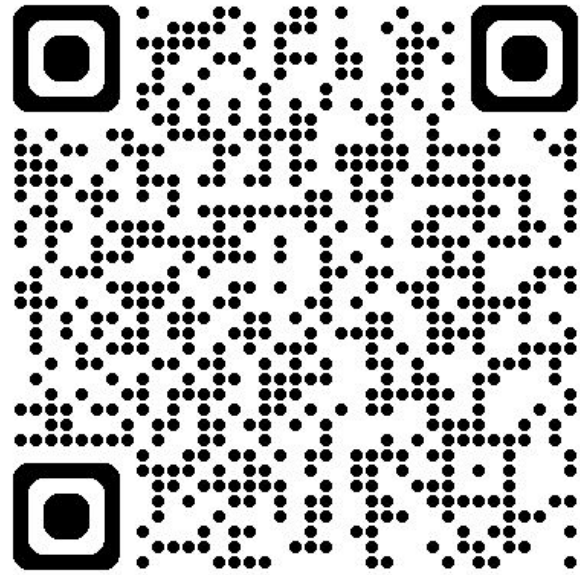
Summary

- Import OpenAPI spec in seconds
- Visualize responses dynamically
- Automate token handling
- Use scripts + variables to accelerate tests

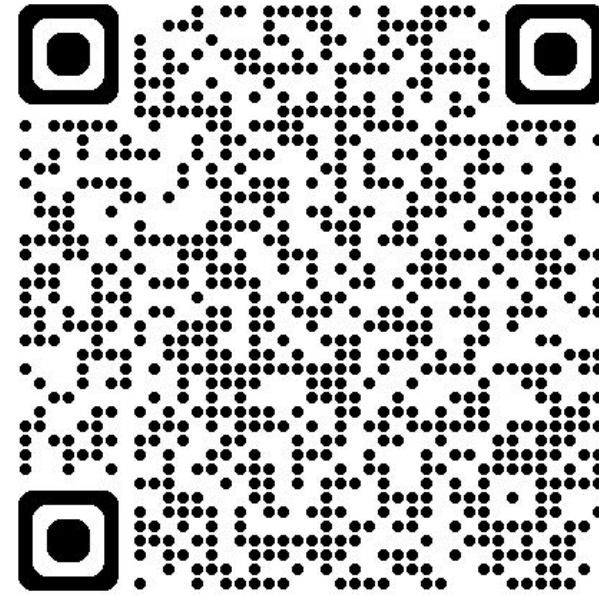
CONGRATS You just learned how to go from Spec to Visuals and accelerate Rapid API Testing with Postman!



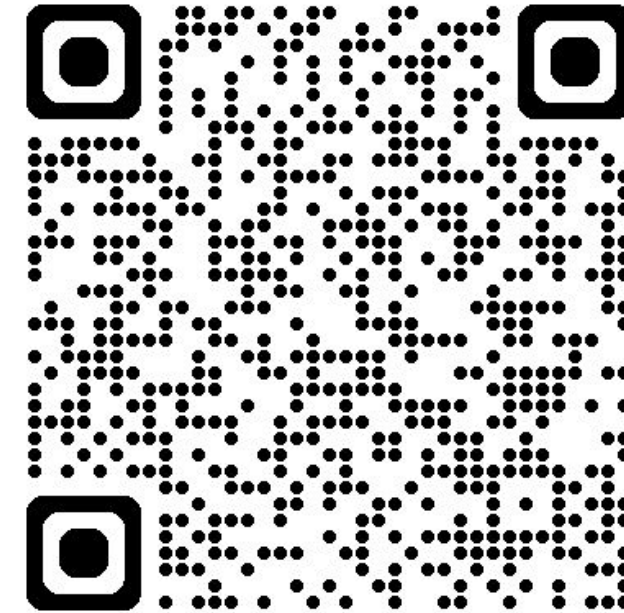
Bonus



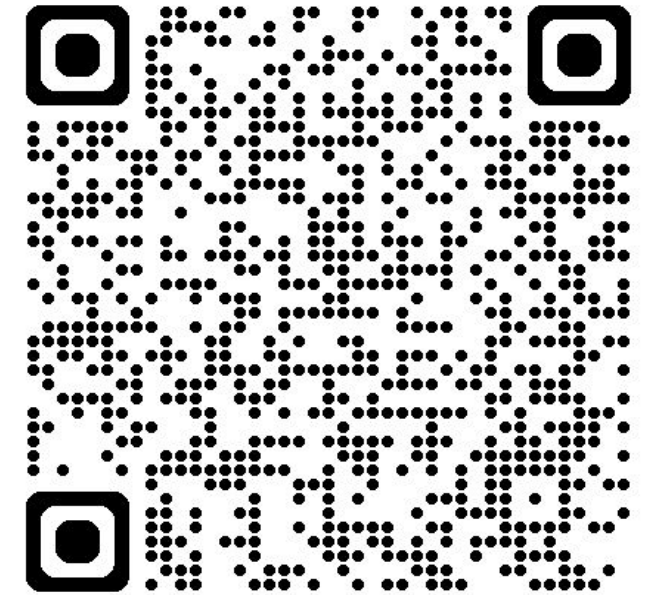
The True Cost
of API Chaos



Generate Run reports with
Newman



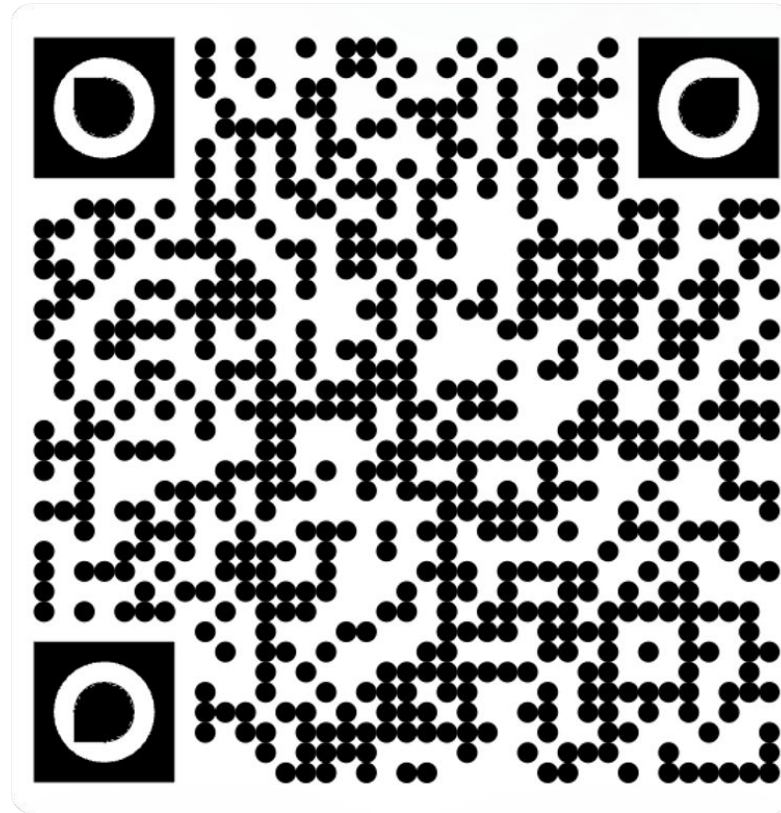
Visualize Request Responses and
write prerequisite scripts





THANK YOU SO MUCH

until next time...



Connect with me

